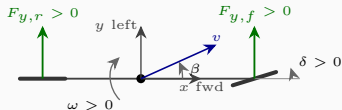


(a) Vehicle body frame (FLU)



$$\alpha_f = \delta - \arctan \frac{v_y + l_f \omega}{v_x}, \quad \alpha_r = -\arctan \frac{v_y - l_r \omega}{v_x}$$

Positive α produces positive (leftward) F_y . Wheel order in every per-wheel array: [FL, FR, RL, RR].

(b) Conversions applied by SimBridge

Channel	Bridge conversion
/odom pose	$y \mapsto -y; \psi, \theta \mapsto -\psi, -\theta$
/odom twist	world $\rightarrow$ body by q^{-1} ; v_y, ω_y, ω_z negated
feedback/imu	$a_y \mapsto -a_y$
tire_forces α	deg $\rightarrow$ rad only: the frame flip and CARLA's own sign convention cancel
tire_forces F_y	$F_y \mapsto -F_y$, into $+y$ left
/drive δ	negated on the way out to CARLA